

```
1 =====
2   EVENT MANAGEMENT APP – DEPLOYMENT GUIDE (Create React App)
3   Frontend Port: 3000 | Backend Port: 5000
4 =====
5
6 -----
7 STEP 1: LAUNCH EC2 INSTANCE
8 -----
9 - Go to AWS Console → EC2 → Launch Instance
10 - Choose Ubuntu (22.04 LTS recommended)
11 - Instance type: t2.micro (free tier) or higher
12 - Key pair: Create or select existing (.pem file)
13 - Keep default storage (8GB is fine)
14 - Launch the instance
15
16 -----
17 STEP 2: CREATE SECURITY GROUP (Inbound Rules)
18 -----
19 Add the following inbound rules:
20
21   Type          Protocol  Port    Source
22   -----
23   SSH           TCP       22      0.0.0.0/0
24   HTTP          TCP       80      0.0.0.0/0
25   Custom TCP    TCP      5000    0.0.0.0/0  ← Backend
26   Custom TCP    TCP      3000    0.0.0.0/0  ← Frontend (CRA)
27
28 NOTE: Use 0.0.0.0/0 for all CIDR blocks.
29 Attach this security group to your instance.
30
31 -----
32 STEP 3: CONNECT TO YOUR INSTANCE VIA SSH
33 -----
34 chmod 400 your-key.pem
35 ssh -i your-key.pem ubuntu@YOUR_VM_IP
36
37 -----
38 STEP 4: UPDATE SYSTEM & INSTALL NODE + NPM
39 -----
40 sudo apt update -y
41 sudo apt upgrade -y
42 sudo apt install nodejs npm -y
43
44 Verify installation:
45 node -v
46 npm -v
47
48 NOTE: If Node version is too old, install via NodeSource:
```

```
49     curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
50     sudo apt install -y nodejs
51
52     -----
53 STEP 5: CLONE THE REPOSITORY
54     -----
55 git clone https://github.com/kaustubhgharat/Event_Management
56 cd ~/Event_Management
57
58     -----
59 STEP 6: CONFIGURE FRONTEND ENV FILE
60     -----
61 nano ~/Event_Management/frontend/.env
62
63     Change this line:
64     REACT_APP_API_URL=http://localhost:5000
65
66     To:
67     REACT_APP_API_URL=http://YOUR_VM_IP:5000
68
69 Save: Ctrl + O → Enter → Ctrl + X
70
71 IMPORTANT: Replace YOUR_VM_IP with your actual EC2 public IP address.
72
73     -----
74 STEP 7: START BACKEND
75     -----
76 cd ~/Event_Management/backend
77 npm install
78 node app.js
79
80 OPTIONAL – Use nodemon for auto-restart on file changes:
81     npm install -g nodemon
82     nodemon app.js
83
84 NOTE: Keep this terminal open. Open a new SSH session for frontend.
85
86     -----
87 STEP 8: START FRONTEND (New Terminal / SSH Session)
88     -----
89 cd ~/Event_Management/frontend
90 npm install
91 npm start
92
93 NOTE: CRA runs on port 3000 by default.
94
95     -----
96 STEP 9: TEST IN BROWSER
```

```
97 -----
98 Open: http://YOUR_VM_IP:3000
99
100 -----
101 TROUBLESHOOTING: RAM / Memory Issues
102 -----
103 If npm install or npm start crashes or hangs, add swap space:
104
105     sudo fallocate -l 1G /swapfile
106     sudo chmod 600 /swapfile
107     sudo mkswap /swapfile
108     sudo swapon /swapfile
109     free -h
110
111 OPTIONAL: To make swap permanent across reboots:
112     echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
113
114 TIP: Try running frontend first, then backend. If it still hangs,
115 apply the swap fix above and retry.
116
117 -----
118 TROUBLESHOOTING: Port Already in Use
119 -----
120 If port 5000 or 3000 is busy:
121     sudo lsof -i :5000
122     sudo kill -9 <PID>
123
124 -----
125 TROUBLESHOOTING: npm not found / old version
126 -----
127     node -v      ← should be 16+ ideally
128     npm -v
129     sudo npm install -g npm@latest
130
131 =====
132     QUICK REFERENCE
133 =====
134     Backend URL : http://YOUR_VM_IP:5000
135     Frontend URL: http://YOUR_VM_IP:3000
136     .env file   : ~/Event_Management/frontend/.env
137     Env variable: REACT_APP_API_URL
138 =====
139
```